Locking Away Prosperity? Evaluating the Labor Impacts of Vaccine Mnadates

1 Introduction

The COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs) as governments acted swiftly to limit viral transmission. Much early research focused on the public health rationale for these policies—especially the importance of early action to contain exponential spread. For instance, Friedson et al. [2021] and Yang [2021] use synthetic control methods to evaluate lockdowns in California and Wuhan, emphasizing the critical role of timing. Similarly, Bayat et al. [2020] estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown occurred just ten days earlier. Subsequent work examined the economic consequences of such interventions. While NPIs helped flatten the curve, they also raised concerns about business closures, layoffs, and lasting economic scarring. Studies like Gibson and Sun [2023] and Lee and Yang [2022] document labor market costs of lockdowns, while Ke and Hsiao [2022] show that although strict NPIs incur short-term economic losses, those effects may be temporary.

By 2021, as vaccines became widely available, cities such as New York City (NYC), New Orleans (NOLA), Los Angeles (LA), and San Francisco (SF) turned to vaccine mandates as a less restrictive (but still controversial) tool for curbing transmission.[1]. Unlike lockdowns, mandates aimed to sustain economic activity in industry while reducing health risks in high-contact settings. Nevertheless, critics argued that requiring proof of vaccination for indoor dining and entertainment could slow employment recovery in restaurants by alienating customers or exacerbating staffing shortages [Dive, 2021]. Supporters countered that mandates would improve public safety and restore consumer confidence, potentially boosting demand and employment [Jones, 2021].

Each city's mandate differed in timing and design. NYC's *Key to NYC* program, announced in August 2021 and enforced beginning September, was the first large-scale vaccine mandate for indoor commerce in the United States [Fortney, 2021]. San Francisco followed with an even stricter rule requiring full vaccination for customers and employees, effective later that same month [Duffett, 2021]. New Orleans adopted a hybrid approach, permitting either proof of vaccination or a recent negative test [Lorell, 2021], while Los Angeles—by far the largest jurisdiction—implemented its policy in November [Reyes, 2021]. These mandates created a

[1] Henceforth, I refer to these simply as the treated units

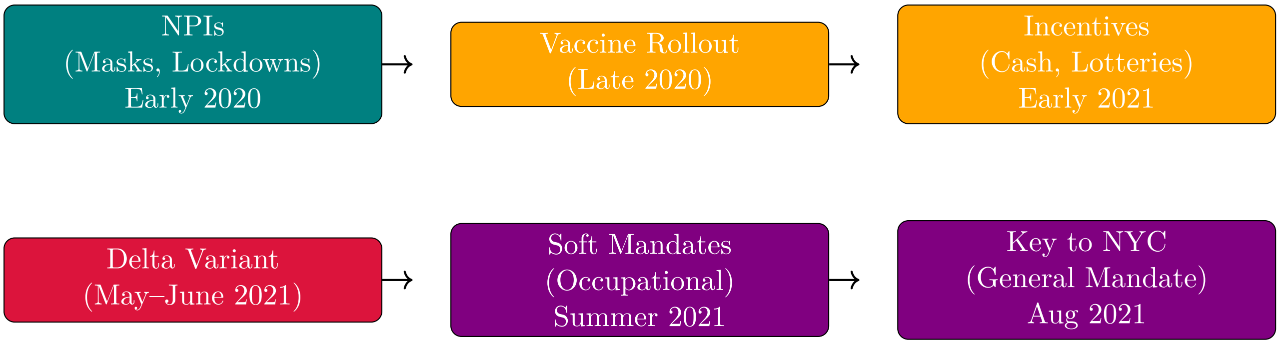
small but consequential set of urban outliers in the national policy landscape, with important implications for labor markets in the leisure and hospitality sector. Precisely, the estimated effects of city-level vaccine mandates on restaurant employment suggest several potential lessons for policymakers and business stakeholders. If the mandates support employment growth, policymakers could view them as a viable tool for balancing public health and economic activity. Cities considering similar interventions might adopt comparable policies with confidence that labor markets will not be severely disrupted. Businesses could plan for future public health policies with reduced concern for employment losses in high-contact sectors. If the mandates depress employment, policymakers should weigh the trade-offs more carefully before implementing mandates, especially in sectors sensitive to consumer demand and staffing shortages. Alternative interventions, such as targeted incentives for vaccination, improved workplace safety measures, or temporary subsidies, might be preferable to blunt employment impacts of mandates. Businesses may need mitigation strategies, including flexible staffing or remote services, to reduce vulnerability to future mandates. If the effects are heterogeneous across cities, differences in local conditions such as compliance rates, labor market tightness, or enforcement intensity should guide tailored policy decisions rather than blanket mandates. Comparative case studies could help identify which contextual factors drive positive versus negative outcomes.

This paper evaluates the causal effects of these city-level vaccine mandates on restaurant employment. While prior research has analyzed mandates' impacts on vaccination rates and health outcomes [Cohn et al., 2022, Acharya and Dhakal, 2021, Lang et al., 2023], relatively little is known about their labor market consequences, especially in industries dependent on face-to-face interaction. This study helps fill that gap. Because randomized experiments are infeasible, the analysis adopts a synthetic control approach to estimate the counterfactual trajectory of restaurant employment absent the mandates. Using MSA-level monthly data from the Federal Reserve Bank of St. Louis, I construct synthetic controls for each mandate city's year-over-year growth in full-service restaurant employment. The donor pool consists of metropolitan areas that did not implement similar mandates during the same period.[2]

The rest of the paper is organized as follows. Section 2 provides background on the evolution of pandemic interventions and the city-level vaccine mandates. Section 3 describes the employment data and outcome construction. Section 4 outlines the empirical strategy, including the synthetic control methodology. Section 5 presents the main findings and robustness checks. Section 6 interprets the results and discusses limitations. Section 7 concludes.

[2]For example, 'SMU11000007072251101SA' (Washington, DC) or 'SMU08197407072251101SA' (Denver MSA).

2 Policy Background



Timeline of major pandemic policy phases

2.1 NPIs Before Vaccines

NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. The first and most famous of these interventions was the 2020 lockdown in Wuhan, China, where local officials went as far as to forcibly seal people in their homes to reduce the possibility to spread the virus; similar actions were taken in Europe and India. Even in the United States, states that could effectively seal their borders, such as Hawaii, did so, with Hawaii mandating two-week quarantines to visitors, which effectively amounted to a tourism ban. Mask mandates were widely adopted as well due to their low cost and direct mechanism of action [Hansen and Mano, 2023, Chernozhukov et al., 2021, Mitze et al., 2020]. While not as severe as lockdowns, the idea of mask mandates was to allow for some freedom of movement while minimizing the chance to spread the virus further. While both sets of policies were effective, they provoked political and economic backlash due to their disruptive nature.

2.2 The First Vaccine Policies

The arrival of vaccines in late 2020 ushered in a new phase of pandemic governance, giving jurisdictions a wider array of policy options to employ. Now, instead of lockdowns, governments finally had pharmaceutical interventions in their toolkit. For the first few months, it worked: when vaccines were released, initial uptake was high, particularly among older adults and (often mandated for) healthcare workers. However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination [Barber and West, 2022, Robertson et al., 2021]. These included direct cash payments, vaccine lotteries (e.g., Ohio's "Vax-a-Million" from May to June [Fuller et al., 2022]), and smaller symbolic giveaways. One paper dubbed this the era of "donuts, drugs, booze, and guns," reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements [Tinari and Riva, 2021].

Despite their creativity, most incentives had modest or short-lived effects [Saban et al., 2021, Walkey et al., 2021, Khazanov et al., 2023]. Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people. Given a new more contagious strain, soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment [American Medical Association et al., 2021]. Over the summer, local governments and organizations expanded this approach [American Medical Association et al., 2021, Al Jazeera, 2021].

2.3 City-Level Vaccine Mandates

The summer 2021 Delta surge spurred several major U.S. cities to enact vaccine verification requirements for indoor restaurants and leisure venues. Each city's approach differed slightly in timing and scope, but all shared the common goal of linking access to indoor leisure activities with proof of vaccination. Together, the treated units formed the clearest urban experiments in the use of vaccine mandates to manage both public health risks and consumer confidence in the hospitality sector.

New York City's "Key to NYC" program was the first such mandate in the United States. Announced on August 3, 2021, formalized on August 16, and enforced beginning September 13, it required proof of at least one vaccine dose to enter indoor dining, fitness centers, and entertainment venues [Benveniste, 2021]. Importantly, the rule applied not only to patrons but also to employees of covered establishments. This dual requirement meant that the policy had the potential to affect restaurant employment through both the supply side, if workers resisted vaccination or exited the industry, and the demand side, if consumer traffic shifted in response to the mandate. The city threatened fines for noncompliance and required businesses to post signage and maintain written protocols, though the intensity of inspections varied. Large chains tended to comply quickly [Messenger, 2021], while smaller establishments voiced concern about administrative burdens and customer pushback. Several lawsuits challenged the mandate, but city officials argued that rising vaccination rates demonstrated its effectiveness [Wiener-Bronner, 2021]. San Francisco followed closely, enacting its mandate on August 20, 2021. Unlike New York, the city required full vaccination rather than just one dose, and did not allow a testing alternative [San Francisco, 2021, Duffett, 2021]. This made San Francisco's policy the strictest among the four cities. The requirement extended to both customers and employees in indoor restaurants, bars, and entertainment venues. Enforcement included spot checks and the threat of fines, with compliance reported to be widespread among larger businesses in the city's dense urban core. As in New York, officials promoted the policy as a tool for stabilizing consumer confidence in public spaces, even as smaller businesses expressed anxiety about turning customers away.

New Orleans became the first major city in the South to adopt a mandate, announcing its policy on August 16, 2021. The city's approach was a hybrid model: individuals could gain entry to indoor restaurants and venues either by showing proof of at least one vaccine dose or by presenting a recent negative COVID test [Aya and Riess, 2021]. This flexibility reflected local political and cultural dynamics but also introduced more administrative complexity for restaurants. Enforcement began within weeks [Just, 2021a,b], with inspections carried out by city officials. Restaurant owners reported mixed reactions, with some noting relief that the policy might make indoor dining safer, while others expressed concern about deterring unvaccinated tourists and residents [Chavez, 2021]. Los Angeles entered the field later, passing its ordinance in October 2021 with enforcement beginning November 4. The policy was sweeping in scope, covering indoor restaurants, gyms, theaters, salons, and other leisure establishments. By the time it came into effect, public debate had already been shaped by the experience of New York and San Francisco, and Los Angeles became the largest jurisdiction in the nation to implement a vaccine passport [Solis and Branson-Potts, 2021].

Although the four cities differed in timing and strictness, they shared important common features. Each policy was applied citywide, targeted indoor leisure and hospitality settings, and created new compliance responsibilities for businesses. They also diverged sharply from the policy environments of states such as Texas and Florida, which prohibited vaccine requirements altogether. The result was a small cluster of urban outliers using vaccine mandates as a central public health intervention in 2021. These institutional details carry important implications for research design. Because the mandates covered both patrons and employees in most cases, they may have constrained labor supply by excluding some workers while simultaneously influencing labor demand by shaping customer traffic. Enforcement intensity and public perception varied, but each policy represented a reasonably sharp and well-documented intervention, announced and enforced on a specific timeline. This combination makes them suitable for empirical evaluation. Comparing employment outcomes in these cities against synthetic control groups of metropolitan areas without mandates offers a credible way to estimate causal effects on restaurant employment. The staggered adoption across cities and the variation in stringency further enrich the research design, providing multiple natural experiments within a single policy family.

2.4 The First Vaccine Mandates

Per the above, vaccine mandates could influence employment in full-service restaurants through both supply- and demand-side channels, with theoretically ambiguous net effects. On the supply side, some workers may exit the labor force or shift sectors if unwilling to comply, especially in lower-wage service jobs where vaccine hesitancy may be higher. However, others may comply with the mandate precisely *because* they wish to keep their jobs, choosing continued employment over personal objection. That is, the employment effect may hinge less on individual preferences in isolation and more on the strength of the economic incentives at stake. Employers, meanwhile, may face frictional costs from enforcing mandates or accommodating staffing turnover. On the demand side, mandates may increase consumer confidence by signaling a safer environment, thereby encouraging more patrons to return to indoor dining. For a sector like full-service restaurants, where both interpersonal contact and discretionary spending are central, mandates could plausibly stimulate demand even as they introduce short-run supply constraints. Whether employment rises or falls in net depends on which margin dominates. In short, this analysis seeks to quantify the net effect of the mandate on restaurant employment, capturing the balance between these potentially opposing forces.

3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED) and, by extension, the Bureau of Labor Statistics, spanning January 1991 through December 2022, yielding $T = 384$ monthly observations per unit. Let $\mathcal{J} = \{1, \ldots, J\}$ denote the set of treated units, corresponding to New York City, San Francisco, New Orleans, and Los Angeles. For treated unit $j \in \mathcal{J}$, the pre-treatment period is defined as

$$\mathcal{T}_1^{(j)} = \{t \in \mathbb{N} : t \leq T_0^{(j)}\},$$

where $T_0^{(j)}$ is the final period before the mandate in city j. Similarly, the post-treatment period is

$$\mathcal{T}_2^{(j)} = \{t \in \mathbb{N} : t > T_0^{(j)}\}.$$

Our analysis focuses on two related outcome sets: (1) full-service restaurant employment for the treated cities and 30 additional MSAs, and (2) broader leisure industry employment at the MSA level for approximately 255 MSAs from 1991 to December 2022.

Let e_{jt} denote employment in unit j at month t, and define the year-over-year growth rate in percentage points as

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \qquad j = 1, \ldots, N, \ t = 13, \ldots, T,$$

using smoothed and seasonally adjusted values. Outcomes for each unit are collected into vectors $\mathbf{y}_j = (y_{j1}, \ldots, y_{jT})^\top$.

For each treated city $j \in \mathcal{J}$, a separate synthetic control is estimated using a donor pool

$$\mathcal{N}_0^{(j)} = \mathcal{N} \setminus \Big(\{j\} \cup \mathcal{J} \setminus \{j\}\Big),$$

which excludes the treated unit itself as well as the other treated units. The treatment indicator is now

$$d_{jt} = \mathbf{1}\{j \in \mathcal{J} \land t > T_0^{(j)}\},$$

so observed outcomes satisfy

$$y_{jt} = (1 - d_{jt})y_{jt}(0) + d_{jt}y_{jt}(1),$$

with causal effects

$$\tau_{jt} = y_{jt}(1) - y_{jt}(0).$$

The post-treatment average treatment effect on the treated (ATT) is

$$\text{ATT} = \frac{1}{J} \sum_{j \in \mathcal{J}} \frac{1}{|\mathcal{T}_2^{(j)}|} \sum_{t \in \mathcal{T}_2^{(j)}} \tau_{jt},$$

averaging first over the post-treatment months within each city and then across all treated cities.

4 Econometric Methodology

Unfortunately, Standard SCM yielded inadequate pre-treatment fit. In order to generate the counterfactual, I needed to allow extrapolation beyond the convex hull. To estimate the counterfactual employment trajectory for NYC's full-service restaurant sector and the broader employment trends in the leisure labor market, I therefore employ a synthetic control approach that hybridizes Forward Selection SCM (FSCM) and the Augmented SCM (ASCM) estimators. The classical Synthetic Control Method (SCM) solves

$$\mathbf{w}^{\text{SCM}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{conv}}}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2,$$

where

$$\mathcal{W}_{\text{conv}} = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{N_0} \;\middle|\; \mathbf{1}^\top \mathbf{w} = 1 \right\}$$

is the probability simplex. This finds the weights $\mathbf{w}^{\text{SCM}}$ that best reconstruct the treated unit's pre-treatment outcomes as a convex combination of donor outcomes.

4.1 Forward Selection SCM

FSCM constructs the synthetic control iteratively by adding donor units one at a time. Starting with an empty selected donor set $\mathcal{S} = \emptyset$, at each iteration the algorithm evaluates each candidate donor $j \in \mathcal{N}_0 \setminus \mathcal{S}$ by forming an expanded set $\mathcal{S}' = \mathcal{S} \cup \{j\}$. For each $\mathcal{S}'$, it solves the restricted SCM problem over the donor submatrix $\mathbf{Y}_0^{\mathcal{S}'}$, consisting of columns corresponding to units in $\mathcal{S}'$:

$$\mathbf{w}_{\mathcal{S}'}^{*} = \underset{\mathbf{w} \in \mathcal{W}_{\text{conv}}(\mathcal{S}')}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1, \mathcal{S}'} \mathbf{w} \right\|_2^2,$$

where

$$\mathcal{W}_{\text{conv}}(\mathcal{S}') = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{|\mathcal{S}'|} \mid \mathbf{1}^\top \mathbf{w} = 1 \right\}.$$

The donor selected at iteration k is

$$j^* = \underset{j \in \mathcal{N}_0 \setminus \mathcal{S}}{\operatorname{argmin}} \min_{\mathbf{w} \in \mathcal{W}_{\text{conv}}(\mathcal{S} \cup \{j\})} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1, \mathcal{S} \cup \{j\}} \mathbf{w} \right\|_2^2,$$

and $\mathcal{S}$ is updated as

$$\mathcal{S} \leftarrow \mathcal{S} \cup \{j^*\}.$$

This forward selection continues until a stopping criterion is met, such as reaching a maximum number of donors, a minimum improvement threshold, or an information criterion like modified BIC.

4.2 The Augmented Synthetic Control Estimator

Building on the baseline SCM, ASCM relaxes the convex weight restriction by allowing affine weights and adds a regularization penalty that shrinks the solution towards a reference weight vector $\mathbf{w}_0$, typically the SCM or FSCM solution. The augmented objective is

$$\mathbf{w}^{\text{aug}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}_0 \right\|_2^2,$$

where $\lambda \geq 0$ controls the penalty strength and

$$\mathcal{W}_{\text{aff}} = \left\{ \mathbf{w} \in \mathbb{R}^{N_0} \mid \mathbf{1}^\top \mathbf{w} = 1 \right\}$$

is the affine subspace of weights summing to one, but not restricted to be non-negative. The intuition is that when $\mathbf{w}_0$ (e.g., FSCM weights) provides a good fit, λ will be large to maintain interpretability; when fit needs improvement, λ can be small to allow extrapolation beyond the convex hull.

4.3 Forward Augmented Synthetic Control

FASC integrates the interpretability of FSCM with the flexibility of ASCM by first performing forward selection to obtain $\mathbf{w}^{\text{FSCM}}$, then solving

$$\mathbf{w}^{\text{FASC}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}^{\text{FSCM}} \right\|_2^2,$$

where $\lambda \geq 0$ balances fidelity to the sparse forward-selected weights and pre-treatment fit.

Large λ yields a solution close to $\mathbf{w}^{\text{FSCM}}$, preserving sparsity and interpretability, whereas small λ prioritizes fit, allowing positive or negative weights outside the original selected donors.

4.3.1 Cross-Validation for λ

The hyperparameter λ is chosen by time-split cross-validation over the pre-treatment period. The training set (first half of pre-treatment) is used to estimate

$$\mathbf{w}^{\text{FASC}}(\lambda) = \underset{\mathbf{w} \in \mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \|\mathbf{y}_{\text{train}} - \mathbf{Y}_{0,\text{train}}\mathbf{w}\|_2^2 + \lambda \left\|\mathbf{w} - \mathbf{w}^{\text{FSCM}}\right\|_2^2,$$

while the validation set (second half of pre-treatment) evaluates out-of-sample error via

$$\text{CV-RMSE}(\lambda) = \left\|\mathbf{y}_{\text{val}} - \mathbf{Y}_{0,\text{val}}\mathbf{w}^{\text{FASC}}(\lambda)\right\|_2.$$

The λ minimizing CV-RMSE balances interpretability and fit. Bayesian optimization over $\log_{10}(\lambda) \in [-2, 3]$ with Gaussian Process surrogates and Expected Improvement acquisition efficiently explores this space. Since $\mathbf{w}^{\text{FSCM}}$ is feasible in the FASC optimization, the in-sample Mean Squared Error (MSE) satisfies

$$\text{MSE}^{\text{FASC}} = \frac{1}{T_0}\left\|\mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1}\mathbf{w}^{\text{FASC}}\right\|_2^2 \leq \frac{1}{T_0}\left\|\mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1}\mathbf{w}^{\text{FSCM}}\right\|_2^2 = \text{MSE}^{\text{FSCM}},$$

ensuring FASC's fit is never worse and typically better than FSCM.

5 Results

6 Discussion

7 Conclusion

References

Binod Acharya and Chandra Dhakal. Implementation of state vaccine incentive lottery programs and uptake of covid-19 vaccinations in the united states. *JAMA Network Open*, 4(12):e2138238–e2138238, 12 2021. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2021.38238.

Al Jazeera. Covid-19 delta surge in us leads to new restrictions, jab push, July 2021. URL https://tinyurl.com/yhbl97yx. Officials are growing more worried about the current rise in infections and are blaming the unvaccinated for the spike.

American Medical Association et al. Major health care professional organizations call for covid-19 vaccine mandates for all health workers. https://tinyurl.com/24r6jjtv, July 2021. Press Release, July 26, 2021.

Elamroussi Aya and Rebekah Riess. New orleans will require proof of vaccination or a negative covid-19 test to enter certain indoor venues, 2021. URL https://tinyurl.com/3wa6cn5h. Accessed: 2025-09-29.

Andrew Barber and Jeremy West. Conditional cash lotteries increase covid-19 vaccination rates. *Journal of health economics*, 81:102578, 2022.

Niloofar Bayat, Cody Morrin, Yuheng Wang, and Vishal Misra. Synthetic control, synthetic interventions, and covid-19 spread: Exploring the impact of lockdown measures and herd immunity, 2020. URL https://arxiv.org/abs/2009.09987.

Alexis Benveniste. New york city will require vaccines for entry to restaurants and gyms, 2021. URL t.ly/Q6Gus.

Roby Chavez. As new orleans parades out vaccine rules, restaurants wonder if they'll make it to mardi gras, 2021. URL https://tinyurl.com/yw974mr4.

Victor Chernozhukov, Hiroyuki Kasahara, and Paul Schrimpf. Causal impact of masks, policies, behavior on early covid-19 pandemic in the u.s. *Journal of Econometrics*, 220(1): 23–62, 2021. doi: 10.1016/j.jeconom.2020.09.003.

Ezra Cohn, Michael Chimowitz, Theodore Long, Jay K Varma, and Dave A Chokshi. The effect of a proof-of-vaccination requirement, incentive payments, and employer-based mandates on covid-19 vaccination rates in new york city: a synthetic-control analysis. *The Lancet Public Health*, 7(9):e754–e762, 2022.

Restaurant Dive. San francisco restaurants blame sales declines on vaccine mandates and delta surge, 2021. URL https://www.restaurantdive.com/news/sf-restaurants-sales-decline-vaccine-mandates. Accessed: 2025-09-29.

Becky Duffett. Sf officially mandates proof of vaccination for indoor dining, 2021. URL t.ly/qCKkk. Accessed: 2025-09-29.

Luke Fortney. Nyc to require proof of vaccination to dine indoors starting august 16, 2021. URL t.ly/jYNQe. Accessed: 2025-09-29.

Andrew I. Friedson, Drew McNichols, Joseph J. Sabia, and Dhaval Dave. Shelter-in-place orders and public health: Evidence from california during the covid-19 pandemic. *Journal of Policy Analysis and Management*, 40(1):258–283, 2021. doi: 10.1002/pam.22267.

Sam Fuller, Sara Kazemian, Carlos Algara, and Daniel J. Simmons. Assessing the effectiveness of covid-19 vaccine lotteries: A cross-state synthetic control methods approach. *PLOS ONE*, 17(9):1–18, 09 2022. doi: 10.1371/journal.pone.0274374.

John Gibson and Xiaojin Sun. A synthetic control analysis of us state level covid-19 stay-at-home orders on new jobless claims. *Journal of Economics and Finance*, 47(1):1–14, 2023.

Niels-Jakob H. Hansen and Rui C. Mano. Mask mandates save lives. *Journal of Health Economics*, 88:102721, 2023. ISSN 0167-6296. doi: https://doi.org/10.1016/j.jhealeco.2022.102721. URL https://www.sciencedirect.com/science/article/pii/S0167629622001357.

Alexandra Jones. Restaurants need workers. would an employee vaccine mandate bring them back?, 2021. URL t.ly/8D796. Accessed: 2025-09-29.

Amie Just. Saints fans at superdome must show covid vaccine proof; here's how it will work, 2021a. URL t.ly/_dS5o.

Amie Just. Thousands sign up for la wallet app after new orleans announces vaccine mandate, 2021b. URL https://tinyurl.com/ye9s98pb.

Xiao Ke and Cheng Hsiao. Economic impact of the most drastic lockdown during covid-19 pandemic—the experience of hubei, china. *Journal of Applied Econometrics*, 37(1):187–209, 2022. doi: 10.1002/jae.2871.

Gabriela K. Khazanov, Rebecca Stewart, Matteo F. Pieri, Candice Huang, Christopher T. Robertson, K. Aleks Schaefer, Hansoo Ko, and Jessica Fishman. The effectiveness of financial incentives for covid-19 vaccination: A systematic review. *Preventive Medicine*, 172:107538, 2023. doi: 10.1016/j.ypmed.2023.107538.

David Lang, Lief Esbenshade, and Robb Willer. Did ohio's vaccine lottery increase vaccination rates? a pre-registered, synthetic control study. *Journal of Experimental Political Science*, 10(2):242–260, 2023. doi: 10.1017/XPS.2021.32.

Jongkwan Lee and Hee-Seung Yang. Pandemic and employment: evidence from covid-19 in south korea. *Journal of Asian economics*, 78:101432, 2022.

Clair Lorell. New orleans mandates proof of vaccination for indoor dining and drinking, 2021. URL t.ly/mBKcc. Accessed: 2025-09-29.

Haley Messenger. From mcdonald's to goldman sachs, here are the companies mandating vaccines for all or some employees, 2021. URL t.ly/3NPwC.

Timo Mitze, Reinhold Kosfeld, Johannes Rode, and Klaus Wälde. Face masks considerably reduce covid-19 cases in germany. *Proceedings of the National Academy of Sciences*, 117 (51):32294–32301, 2020. doi: 10.1073/pnas.2015954117.

Emily Alpert Reyes. L.a. to require proof of covid vaccination at indoor restaurants, salons, other venues, 2021. URL t.ly/dN8-1. Accessed: 2025-09-29.

Christopher Robertson, K Aleks Schaefer, and Daniel Scheitrum. Are vaccine lotteries worth the money? *Economics Letters*, 209:110097, 2021.

Mor Saban, Vicki Myers, Shani Ben Shetrit, and Rachel Wilf-Miron. Issues surrounding incentives and penalties for covid-19 vaccination: The israeli experience. *Preventive Medicine*, 153:106763, 2021. ISSN 0091-7435. doi: https://doi.org/10.1016/j.ypmed.2021.106763. URL https://www.sciencedirect.com/science/article/pii/S0091743521003327.

San Francisco. San francisco to require proof of vaccination for entry to certain indoor businesses and all large indoor events, 2021. URL https://shorturl.at/urolj. Accessed: 2025-09-29.

Nathan Solis and Hailey Branson-Potts. Thousands protest covid-19 vaccination mandates as l.a.'s verification rules kick in, 2021. URL https://tinyurl.com/4s4zw8wa.

Serena Tinari and Catherine Riva. Donuts, drugs, booze, and guns: what governments are offering people to take covid-19 vaccines. *BMJ*, 374, 2021. doi: 10.1136/bmj.n1737. URL https://www.bmj.com/content/374/bmj.n1737.

Allan J. Walkey, Anica Law, and Nicholas A. Bosch. Lottery-based incentive in ohio and covid-19 vaccination rates. *JAMA*, 326(8):766–767, 08 2021. doi: 10.1001/jama.2021.11048.

Danielle Wiener-Bronner. Restaurant owners sue new york city over vaccine mandate, 2021. URL t.ly/5KXac.

Xiaoxuan Yang. Does city lockdown prevent the spread of covid-19? new evidence from the synthetic control method. *Global Health Research and Policy*, 6:1–14, 2021.